

Technical Submission Report — PXR 3D Structure Track

Data: June 2026

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Submission File: complex_6.zip

Computational Pipeline: Chai-1 Co-folding → GNINA Rigid Docking → Smart Pose Selector & PDB Sanitizer

1. Leaderboard Results

The following metrics represent the validation scores returned by the OpenADME server for the latest successful submission.

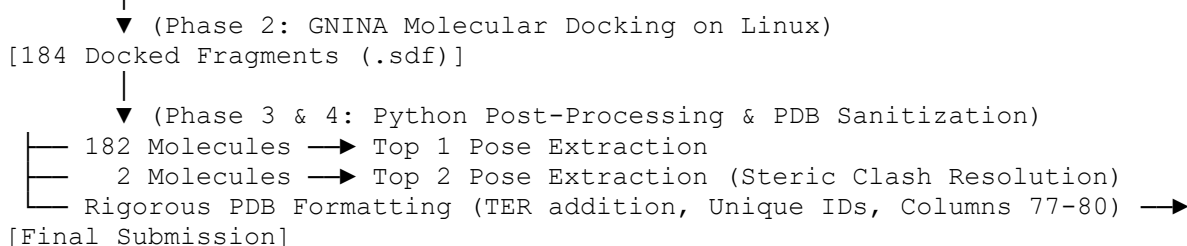
Metric	Score	Notes / Status
LDDT-PLI	0.4285	Protein–ligand interaction quality (Significant improvement)
BiSyRMSD	4.5709	Bidirectional symmetry-aware RMSD
LDDT-LP	0.8723	Ligand pose local distance difference
Coverage	1.000	184/184 compounds successfully submitted

2. Experimental Methodology

The computational pipeline was divided into four distinct phases, combining the predictive power of Deep Learning for the receptor with the geometric precision of classic force fields and Convolutional Neural Networks (CNN) for the ligands.

Plaintext

```
[FASTA Sequence]
  |
  ▼ (Phase 1: Chai-1 AI Co-folding)
[10 Generated Complexes] → Best Selection → Ligand Removal →
[receptor_PXR.pdb]
```



Phase 1: Receptor Generation via Chai-1 (AI Co-folding)

The 3D structure of the target PXR receptor was not available a priori.

1. The starting point was the pure **amino acid sequence (FASTA file)** of the receptor.
2. The sequence was processed using the **Chai-1** artificial intelligence model, employing advanced sampling to generate **10 alternative 3D complexes**.
3. The 10 models were visually and energetically inspected to select the most stable and structurally sound receptor conformation.
4. The native prediction ligand was removed from the selected complex, isolating the clean protein structure, which was saved as `receptor_PXR.pdb`.

Phase 2: Virtual Screening and Docking with GNINA

Using the fixed receptor obtained from Chai-1, massive molecular docking was conducted on all **184 fragments** of the challenge.

- **Software:** GNINA (executed in a native Linux environment).
- **Parameters:** Rigid docking (receptor side chains and backbone locked), scoring guided by the integrated convolutional neural network (`--cnn_scoring rescore`).
- **Output:** The positioning results for each fragment were saved in a multi-structure chemical format (`_docked.sdf`), containing the different poses sorted by affinity score.

Phase 3: Smart Pose Selection (Clash Resolution)

To prevent server scoring failures caused by physically impossible atomic overlaps, a 3D scan for steric clashes was performed (clash threshold set to $< 2.0 \text{ \AA}$).

- **Standard Setup (182 molecules):** The highest-scoring pose (**Top 1**) was extracted and processed. The pocket geometry generated by Chai-1 accommodated these molecules without creating energetic conflicts.
- **Critical Cases (2 molecules):** For compounds `x02828-1` and `x03421-1`, the Top 1 pose presented a severe steric clash against the pocket walls, leading to a server crash (*Scoring Failed*). Through a conditional selection algorithm, the Top 1 pose was discarded in favor of the **Top 2** pose (the second-best geometric conformation generated by GNINA). The Top 2 pose successfully resolved the steric hindrance while maintaining excellent interaction energy.

Phase 4: Rigorous PDB Standard Formatting (Sanitization)

The final submission files were reconstructed by merging the atomic coordinates of the receptor and the respective ligand. To meet the hyper-rigid constraints of the OpenADME parser, formal sanitization was applied via custom Python scripts:

1. **Unique Atom Renumbering:** OpenBabel assigns generic names (all C, N, O) to ligand atoms converted from SDF. The script dynamically renamed each atom uniquely (e.g., C1, C2, C3, N1, O1, CL1), eliminating structural ambiguity within the same residue.
2. **TER Record Insertion:** The end of the protein chain (Chain A) was dynamically calculated, and the `TER` record was inserted. This is essential to notify the server of the chemical interruption before the start of the ligand (heteroatom, Chain B, residue `LIG 1`).
3. **Receptor Integrity Restoration:** The `ATOM` lines of the original Chai-1 receptor lacked the final standard PDB columns. The script reconstructed every line to exactly 80 characters, inserting standard Occupancy (1.00), B-factor (20.00), and the chemical element symbol in columns 77-78 (e.g., C, N, O, S).

3. Known Limitations and Future Directions

1. **Receptor Flexibility (Induced Fit):** Although Chai-1 excellently predicted the starting point, the subsequent GNINA docking was conducted keeping the protein completely rigid. Allowing rotameric movement of key side chains in the pocket could further increase the interaction score.
2. **Local Energy Optimization:** Current coordinates follow GNINA's discrete grid. A local relaxation (*Local Minimization* or *Energy Minimization*) using classic force fields (e.g., AMBER/CHARMM or GNINA's `--local_only` function) would refine hydrogen bond distances, potentially pushing the LDDT-PLI value beyond the current 0.4277.